WORKED SOLUTION

STEP II 2003, Question 2

Pure • Inverse Trigonometric Functions

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/03-s2-q2

The key idea

The expression $a\cos\theta + b\sin\theta$ can always be written as a single cosine, $R\cos(\theta - \alpha)$. Doing that turns the equation into a statement about arccos, while α itself is an arctan. So once we spot a nice solution θ, the identity for π is just "$\theta = (\theta - \alpha) + \alpha$" rewritten. The only real care needed is in checking that each angle lies in the range of the inverse function we use.

Solution

The first identity

Try the obvious angles in $(\frac{1}{4}\pi, \frac{1}{2}\pi)$. With $\theta = \frac{1}{3}\pi$,

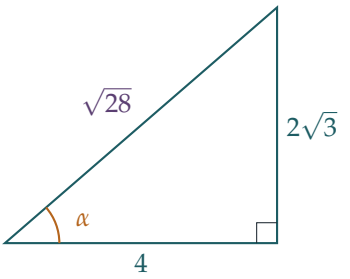
$$4\cos\tfrac{\pi}{3} + 2\sqrt{3}\,\sin\tfrac{\pi}{3} = 4\cdot\tfrac{1}{2} + 2\sqrt{3}\cdot\tfrac{\sqrt{3}}{2} = 2 + 3 = 5,$$

so $\theta = \frac{1}{3}\pi$ works, and $\frac{1}{4}\pi < \frac{1}{3}\pi < \frac{1}{2}\pi$.

Now write the left-hand side in the form $R\cos(\theta - \alpha) = R\cos\theta\cos\alpha + R\sin\theta\sin\alpha$. Matching coefficients,

$$R\cos\alpha = 4, \qquad R\sin\alpha = 2\sqrt{3},$$

so $R^2 = 16 + 12 = 28$, giving $R = \sqrt{28}$, and $\tan\alpha = \frac{2\sqrt{3}}{4} = \frac{\sqrt{3}}{2}$. Since $\cos\alpha$ and $\sin\alpha$ are both positive we may take α acute, so $\alpha = \arctan(\sqrt{3}/2)$.



The equation therefore reads

$$\sqrt{28}\cos(\theta-\alpha)=5, \qquad \text{that is,} \qquad \cos(\theta-\alpha)=\frac{5}{\sqrt{28}},$$

and it is satisfied by $\theta=\frac{1}{3}\pi$.

Ask yourself Can we immediately conclude that $\frac{1}{3}\pi-\alpha=\arccos(5/\sqrt{28})$?

Not quite: $\cos x=c$ has many solutions, and $\arccos c$ is only the one in $[0,\pi]$. So we must check that $\frac{1}{3}\pi-\alpha$ lies in $[0,\pi]$. Since tan is increasing on $(0,\frac{1}{2}\pi)$ and $\tan\alpha=\frac{\sqrt{3}}{2}<\sqrt{3}=\tan\frac{\pi}{3}$, we have $0<\alpha<\frac{1}{3}\pi$. Hence

$$0<\tfrac{1}{3}\pi-\alpha<\tfrac{1}{3}\pi\leq\pi,$$

and so $\frac{1}{3}\pi-\alpha=\arccos(5/\sqrt{28})$. Rearranging and multiplying by 3,

$$\pi=3\arccos(5/\sqrt{28})+3\alpha=3\arccos(5/\sqrt{28})+3\arctan(\sqrt{3}/2),$$

as required.

The second identity

Copy the idea: we want $\frac{1}{4}\pi=\arcsin(7\sqrt{2}/10)-\arctan(3/4)$, so let $\beta=\arctan(3/4)$ and look at $\sin(\frac{1}{4}\pi+\beta)$.

Since $0<\frac{3}{4}$, β is acute, and from the 3–4–5 right-angled triangle, $\sin\beta=\frac{3}{5}$ and $\cos\beta=\frac{4}{5}$. Then, by the addition formula,

$$\begin{aligned}\sin\left(\tfrac{1}{4}\pi+\beta\right)&=\sin\tfrac{\pi}{4}\cos\beta+\cos\tfrac{\pi}{4}\sin\beta\\&=\frac{1}{\sqrt{2}}\left(\frac{4}{5}+\frac{3}{5}\right)=\frac{7}{5\sqrt{2}}=\frac{7\sqrt{2}}{10}.\end{aligned}$$

Ask yourself Is $\frac{1}{4}\pi+\beta$ really the angle that arcsin returns?

The range of arcsin is $[-\frac{1}{2}\pi,\frac{1}{2}\pi]$, so we need $\frac{1}{4}\pi+\beta$ in there. Because $\tan\beta=\frac{3}{4}<1=\tan\frac{\pi}{4}$ and tan is increasing on $(0,\frac{1}{2}\pi)$, we have $0<\beta<\frac{1}{4}\pi$, so

$$\tfrac{1}{4}\pi<\tfrac{1}{4}\pi+\beta<\tfrac{1}{2}\pi.$$

Hence $\arcsin(7\sqrt{2}/10)=\frac{1}{4}\pi+\arctan(3/4)$, and multiplying by 4 and rearranging,

$$\pi=4\arcsin(7\sqrt{2}/10)-4\arctan(3/4),$$

which is the result.

Checks

Numerically, $\arctan(\sqrt{3}/2) \approx 0.713724$ and $\arccos(5/\sqrt{28}) \approx 0.333473$; their sum is $1.047198 \approx \pi/3$, and three times it is 3.141593. For the second identity, $\arcsin(7\sqrt{2}/10) \approx 1.428899$ and $\arctan(3/4) \approx 0.643501$; the difference is $0.785398 \approx \pi/4$. A quick scan also shows that $\frac{1}{3}\pi$ is the only solution of the equation in $(\frac{1}{4}\pi, \frac{1}{2}\pi)$: the other solution near there, $\alpha - \arccos(5/\sqrt{28}) \approx 0.380$, is less than $\frac{1}{4}\pi$.

Common slips

- Writing $\arccos(\cos x) = x$ without checking $x \in [0, \pi]$ (or $\arcsin(\sin x) = x$ without $x \in [-\frac{1}{2}\pi, \frac{1}{2}\pi]$). This is exactly where the marks for rigour are.
- Getting $\tan\alpha$ upside down: from $R\cos\alpha = 4$, $R\sin\alpha = 2\sqrt{3}$ we get $\tan\alpha = 2\sqrt{3}/4$, not $4/(2\sqrt{3})$.

Beyond STEP

Identities expressing π as combinations of inverse tangents are the basis of Machin-type formulae, such as $\frac{1}{4}\pi = 4\arctan\frac{1}{5} - \arctan\frac{1}{239}$, which were used to compute π to many digits because the series for $\arctan x$ converges fast for small x. The clean way to see them is through complex numbers: $\arctan(b/a)$ is the argument of $a + b\mathrm{i}$, so adding inverse tangents corresponds to multiplying complex numbers. Needham's *Visual Complex Analysis* develops this geometric viewpoint beautifully.

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